

SUMMER INTERNSHIP REPORT

ON

Six Laning of Surat – Nashik – Ahmednagar – Solapur –

MH/KNT Border Greenfield Stretch

[Package XIV: km 512.000 (Hasapur) to km 548.400 (Badadal)]

Under Bharatmala Pariyojana

(Hybrid Annuity Mode)

Submitted To: NHAI PIU, Solapur



Submitted by

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Acknowledgement

I would like to express my sincere thanks to everyone who supported me during my internship. I am especially grateful to Mr. Swapnil Kasar, Project Director of NHAI, PIU, Solapur, for giving me the opportunity to be a part of this major highway project. My sincere thanks also go to Mr. Mahesh Yemul, Site Engineer at NHAI, PIU, Solapur, and the rest of the NHAI staff. I highly appreciate their continuous support and coordination, which ensured that my entire internship experience was incredibly smooth and well-facilitated from start to finish.

For the technical knowledge I gained on-site, I want to extend my deepest appreciation to all the engineers and managers of the concessionaire, GR Hasapur Badadal Highway Private Limited. They took the time to answer all my questions, clearly explain the structural drawings, and guide me through the daily construction methodologies. Their practical guidance and cooperative team spirit were the core of my hands-on learning.

This internship gave me valuable practical knowledge that connects perfectly with my civil engineering studies. I truly appreciate the time and effort everyone invested in my training.

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Introduction

This report is a summary of my summer internship which I did at NHAI PIU Solapur. During this internship, I got a chance to work on a big and important road project the Six Laning of the Surat-Nashik-Ahmednagar-Solapur-MH/KNT Border Greenfield Highway under Bharatmala Pariyojana.

The internship helped me understand how a real highway construction project actually works. In college, we read books and solve problems on paper, but on the construction site, everything is different. There are machines, materials, tests, and a lot of teamwork involved.

I worked with many departments during my internship like Quality Control, Safety, Highways, Structures, Surveying, Planning, Billing, Store and Plant. Each department taught me something new and useful.

About NHAI and Bharatmala Pariyojana

NHAI stands for National Highways Authority of India. It is a government body that looks after building and maintaining national highways across the country. PIU Solapur is the Project Implementation Unit that is overseeing this highway project in the Solapur region.

Bharatmala Pariyojana is a big government scheme to build and improve highways all over India. This project is part of that scheme. The mode of contract used here is called Hybrid Annuity Mode (HAM), which means the government pays 40% of the cost during construction and the company gets the rest over 15 years.

About the Concessionaire

GR Infraprojects Limited (GRIL) is one of the well-known infrastructure companies in India. Over the years, GRIL has expanded into railways, metro projects, and even ropeway construction. The company that is directly handling this project is GR Hasapur Badadal Highway Private Limited (GHBHPL), which is a Special Purpose Vehicle (SPV) created specifically for this project.

About Independent Engineer

The Independent Engineer (IE) on this project is M/s Global Infra Solutions JV Voyants Solutions Pvt. Ltd. in association with MS Consultant. The IE is appointed by NHAI and works independently meaning they are not on the side of the contractor and not fully on the side of NHAI either. Their job is to be fair to both parties.

Project Overview

Below is a summary of all the important details about the project I worked on during my internship:

Project Name	Six Laning of Surat–Nashik–Ahmednagar–Solapur–MH/KNT Border Greenfield Stretch [Package XIV from km 512.000 (Hasapur) to km 548.400 (Badadal)], under Bharatmala Pariyojana
Concessionaire (SPV)	GR Hasapur Badadal Highway Private Limited (GHBHPL)
Parent Company	GR Infraprojects Limited
Independent Engineer (IE)	M/s Global Infra Solutions JV Voyants Solutions Pvt. Ltd. in association with MS Consultant
IE Mobilization Date	01-May-2024
Safety Consultant	M/s CDF Infra Con Pvt. Ltd.
Date of Award (LOA)	29-March-2023
Signing of Agreement	06-June-2023
Appointed Date	03-October-2024
Construction Period	730 Days
Total Concession Period	730 Days (Construction) + 15 Years O&M
Scheduled Completion Date	02-October-2026
Scheme / Phase	Bharatmala Pariyojana
Execution Mode	Hybrid Annuity Mode (HAM)
No. of Lanes	6 Lane
Project Length	36.4 km
Total Project Cost	Rs. 872.17 Crore
Major Bridges	1 No.
Minor Bridges	13 Nos.
Rail-Road Bridge (ROB)	1 No.
Vehicular Underpass (VUP/LVUP)	2 Nos. / 9 Nos.
VOP	2 Nos.
Interchange	4 Nos.
Box Culverts / Pipe Culverts	96 / 10 Nos.
Toll Plazas	3 Nos.
Rest Areas	2 Nos.

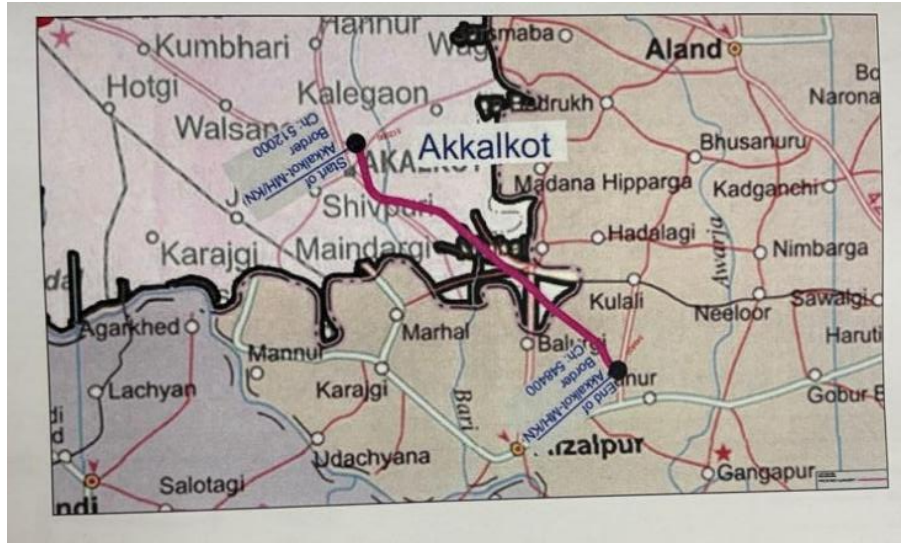


Fig. Location of Road on Map

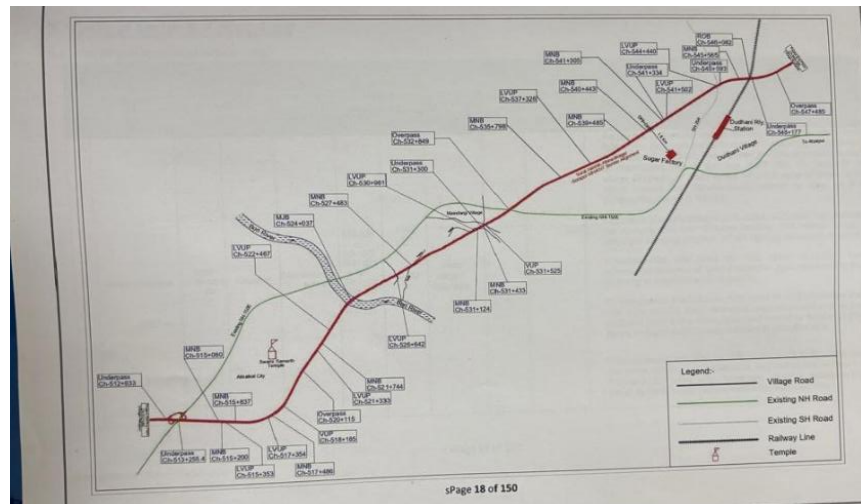


Fig. Map Showing Some Structures Location on Road

Quality Control Department

Quality Control is the department that checks whether the materials and the construction work meet the required standards. On a project like this, if the quality is not good, the road can develop problems like cracks, potholes, or even failures in bridges and structures.

I spent a good amount of time in the QC lab and learned about different tests that are done regularly. Here are some of the tests I understood and observed:

1. Atterberg Limits (Liquid Limit and Plastic Limit) (IS 2720 - Part 5):

- Objective: To determine the plasticity characteristics of the soil, which helps us understand how the subgrade will behave when its moisture content changes.
- Result: Following the IS code procedures, we got a Liquid Limit (LL) of 63%, a Plastic Limit (PL) of 46.04%, and a Plasticity Index (PI) of 16.96%.
- Interpretation: An LL above 50% indicates highly compressible soil, and a PI of ~17% shows moderate plasticity. This means the soil is prone to swelling when in contact with moisture.



Fig. Casagrande Apparatus

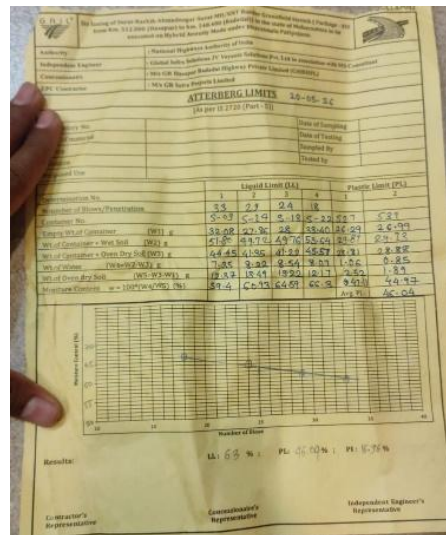


Fig. Results of Atterberg Limit From Lab

2. Free Swell Index Test (IS 2720 - Part 40):

- Objective: To measure the expansion potential of the soil, which is critical for identifying expansive clays that could swell and cause the pavement above to crack or heave when moisture content change.
- Result: Following the IS code procedures, we recorded a final soil volume of 15 ml in distilled water and 11 ml in kerosene, resulting in a Free Swell Index (FSI) of 36%.

- Interpretation: An FSI of 36% indicates a moderate degree of expansiveness. This confirms the soil has a noticeable tendency to swell upon absorbing water, highlighting the need for proper moisture control and stabilized upper layers to protect the road crust.



Fig. Cylinders containing sand filled with water and Kerosene

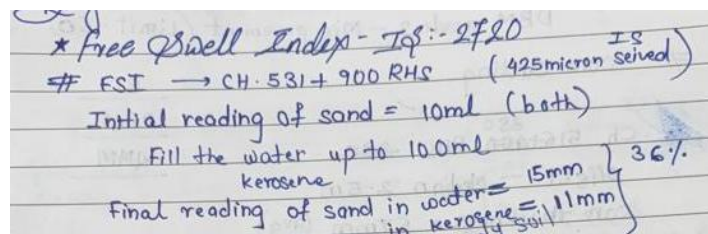


Fig. Results of FSI Test from Lab

3. Modified Proctor's Compaction Test (IS 2720 - Part 8):

- Objective: To determine the Maximum Dry Density (MDD) and Optimum Moisture Content (OMC) of the soil, which is critical for guiding field compaction to achieve maximum structural stability and prevent future pavement settlement.
- Result: Following the IS code procedures compacting the soil in 5 layers with 55 blows each with 4.9 kg rammer then we plotted the compaction curve and recorded a Maximum Dry Density (MDD) of 2.046 g/cc at an Optimum Moisture Content (OMC) of 8.90%.
- Interpretation: This result dictates the exact moisture target for the field team. To achieve the highest strength and load-bearing capacity for the subgrade, the soil must be compacted with exactly 8.90% moisture. Hitting this specific target ensures the subgrade reaches its maximum density of 2.046 g/cc, minimizing air voids and creating a robust, sink-resistant foundation for the upper road crust



Fig. Mould for Test



Fig. Rammer

4. Direct Shear Test (IS 2720 - Part 13):

- Objective: To determine the shear strength parameters of the soil, specifically its cohesion (c) and angle of internal friction (ϕ).
- Result: The test yields the specific c and ϕ values by plotting the shear stress at which the soil sample fails under different normal loads. I observed how the test was carried out on site, though I did not have the exact readings from this particular trial on hand.
- Interpretation: These two parameters are fundamental for calculating the ultimate bearing capacity of the subgrade and evaluating the stability of embankment slopes. A higher angle of internal friction (ϕ) indicates excellent interlocking of granular particles, while cohesion (c) measures the binding force in clayey soils.



Fig. Direct Shear Machine

5. California Bearing Ratio (CBR) Test (IS 2720 - Part 16):

- Objective: To determine the load-bearing capacity of the subgrade soil, which is essential for designing the required thickness of the overlying pavement layers.
- Result: Following the IS code procedure and using the proving ring calibration factor of 3.15, the test on Mould No. 05 gave a CBR value of 37.25% at 2.5 mm penetration (against a standard load of 1370 kg) and 41.69% at 5.0 mm penetration (against a standard load of 2055 kg).
- Interpretation: We use the higher value for the road design, which is 41.69%. This is a very good score. It means the soil from this borrow area is strong and will provide a solid, stable foundation for the road to carry heavy traffic.



Fig. CBR Apparatus

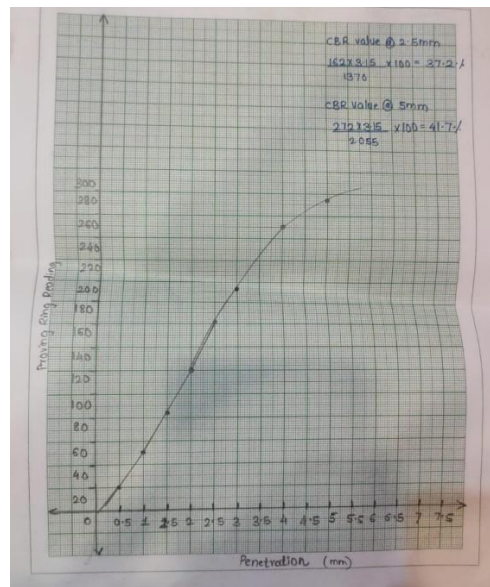


Fig. Graph from Results of CBR Test

6. Softening Point Test (IS 1205):

- Objective: To determine the temperature at which the bitumen binder softens and begins to flow, which is crucial for evaluating its suitability for hot climates and its resistance to temperature-induced pavement deformations.
- Result: Following the Ring and Ball apparatus procedures specified in the IS code, we recorded softening point temperatures of 54.1°C and 54.2°C during our lab trials, successfully passing the minimum requirement of 50°C.
- Interpretation: A softening point in this range confirms that the bitumen is highly suitable for the peak ambient temperatures expected in the project region. It ensures the asphalt binder will remain sufficiently stiff and stable during intense summer heat, effectively preventing the pavement surface from melting, bleeding, or rutting under heavy, continuous traffic loads.



Fig. Apparatus used in Softening point Test



Fig. Rings Filled with Bitumen

7. Penetration Test (IS 1203):

- Objective: To determine the hardness or consistency of the bitumen binder by measuring the depth a standard needle penetrates the sample under a specified load, time, and temperature (typically 100g for 5 seconds at 25°C).
- Result: During our lab trials, we recorded penetration values generally ranging between 45 and 48 units, which safely meets the minimum requirement of 35 for this specific grade.
- Interpretation: The penetration value dictates the grade and stiffness of the bitumen. A lower penetration value indicates a harder, stiffer binder. Using this harder grade is highly intentional for projects in hot climates; it ensures the asphalt wearing courses maintain their structural integrity and firmly resist heat-induced deformations like rutting, shoving, or bleeding under heavy traffic.

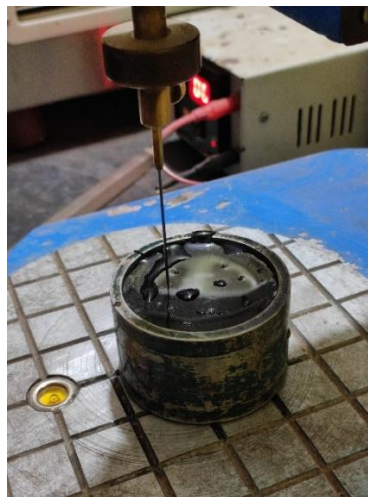


Fig. Needle Penetrating Bitumen

8. Elastic Recovery Test (ASTM D6084):

- Objective: To check if the bitumen can stretch and return to its original shape. This shows how flexible it is.
- Result: The test on the Crumb Rubber Modified Bitumen (CRMB) showed a recovery of 6.3 cm. This gives an elastic recovery of 63%.
- Interpretation: For CRMB, the minimum required limit is 60%. A score of 63% means the material passes easily. It acts like a rubber band. When heavy vehicles drive over the road, this flexibility allows the surface to bend a little and bounce right back. This stops the road from cracking over time.



Fig. Machine used for Elastic Recovery Test



Fig. Mould Casting for the Test

9. Absolute Viscosity Test (IS 1206 - Part 2):

- Objective: To measure how thick and sticky the bitumen is at 60°C. This shows how well it holds up in hot weather.
- Result: Following the procedure as per the IS code, we recorded an average viscosity of 4221.9 Poises.
- Interpretation: The standard limit is between 3200 and 4800 Poises. Since 4221.9 falls inside this range, the bitumen passes the test. This thickness is perfect for hot climates, ensuring the road stays strong and does not melt or form deep tire ruts during summer heat.



Fig. Viscosity Bath Apparatus

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15/05/2024		15/05/2024	
Authority		National Highways Authority of India	
Implementation Engineer		National Highways Authority of India	
Contract Number		NH-48 (Bypass)	

Fig. Results of Absolute Viscosity Test

10. Marshall Stability Test (IS 17127):

- Objective: To measure the maximum load the asphalt mix can hold before it breaks, and how much it bends (flow) under that load.
- Project Result: Testing six samples gave an average dial reading of 368 divisions. Multiplying this by the 0.038 factor, the average Marshall Stability is 13.98 kN. The average flow (settlement) is 3.4 mm.
- Interpretation: A stability of 13.98 kN is a high score. It means the road is very strong and can safely carry heavy trucks. A flow of 3.4 mm is perfectly balanced (standard limits are usually between 2 mm and 4 mm). The mix is stiff enough to prevent deep tire ruts, but flexible enough so it will not crack under traffic.



Fig. Marshall Apparatus

2) Marshall stability → (0.038 multiply)					
1	2	3	4	5	6
390	395	330	405	305	383 → divisions
3.8	3.0	3.0	3.6	3.6	3.4 → settlement (mm)

Fig. Results of Marshall Stability Test

11. Stripping Test - Boiling Method (IS 6241):

- Objective: To check the bond between the bitumen and the stones by visually inspecting them after boiling in water.
- Project Result: Following the procedure as per the IS code, the sample was boiled and then checked visually to see how much bitumen washed off. The visual inspection confirmed that less than 5% of the bitumen coating was removed. This strong bond was achieved using 2% lime and a nanotechnology additive (Zycotherm SP/AS).
- Interpretation: Passing this visual test means the bitumen sticks very tightly to the stones, even under harsh wet conditions. The added lime and chemicals stop water from stripping the bitumen away, which prevents potholes from forming during heavy rain.



Fig. Aggregates after completing Test

12. Fineness of Cement Test (IS 4031 - Part 1):

- Objective: To check how fine the cement powder is by shaking it through a very small net (a 90-micron sieve). This shows how fast the cement will react with water to gain strength.
- Project Result: An initial weight of 100 grams of cement was shaken through a 90-micron sieve. The final weight left on the sieve was 3.3 grams, which gives a fineness value of 3.3%.
- Interpretation: The standard limit requires the leftover weight to be less than 10%. Since 3.3% is well below this limit, the cement passes easily. This means the cement is very fine, will mix well with water, and will help the concrete become strong quickly.

13. Consistency of Cement Test (IS 4031 - Part 4):

- Objective: To find the exact amount of water needed to make a standard cement paste. This is measured by how deep a standard plunger sinks (penetrates) into the paste.
- Project Result: Adding 29% water (101.5 grams) to the 350 g of cement resulted in a plunger penetration of 7 mm from the bottom of the mould.

- Interpretation: The standard requirement is a penetration of 5 mm to 7 mm from the bottom. With a 7 mm reading, the standard consistency of this cement is confirmed as 29%. A normal consistency value is usually between 26% and 33%, so this result is correct. This specific water percentage is now set and will be used for all further setting time and strength tests.



Fig. 7mm Penetration in Cement Paste

14. Initial and Final Setting Time of Cement Test (IS 4031 - Part 5):

- Objective: To find out how long it takes for the wet cement paste to start getting stiff (initial setting) and how long it takes to become completely hard (final setting).
- Project Result: First, the water needed for this test was calculated as 85% of the standard consistency water ($101.5 \text{ gm} \times 0.85 = 86.275 \text{ gm}$). Testing the cement sample gave an initial setting time of 125 minutes, and a final setting time of 235 minutes.
- Interpretation: The standard rule requires the initial setting time to be at least 30 minutes. A result of 125 minutes passes easily, giving workers plenty of time to safely mix and pour the concrete. The final setting time must be less than 600 minutes (10 hours). A result of 3 hours and 55 minutes is very good, meaning the concrete hardens quickly enough so construction can continue without long delays.

15. Individual Gradation of Aggregate Test (IS 2386 - Part 1):

- Objective: To determine the particle size distribution of the stones (coarse aggregates) by shaking them through a series of sieves. This ensures the stones are a good mixture of sizes so they can lock together tightly in the road layers without leaving too many empty gaps.

- Project Result: A 22,470 gm sample was tested. The results showed that 100% passed the 40 mm sieve, 97.56% passed the 20 mm sieve, 4.19% passed the 10 mm sieve, and 0.98% passed the 4.75 mm sieve.
- Interpretation: The aggregate perfectly meets the required specification limits for the 40 mm, 20 mm, and 4.75 mm sieves. There is a very minor deviation at the 10 mm sieve, where 4.19% passed (the standard requirement is between 5% and 20%). This simply means this specific sample had slightly more larger stones (retained on the 10 mm mesh) than the strict ideal limit. Overall, it shows the aggregate is a strong, primarily 20 mm coarse material suitable for the highway layers.

IS Sieve Size (mm)	Weight Retained (g)	Cumulative Weight Retained (g)	Cumulative Retained (%)	Passing (%)	Remark
40	0	0	0	100	100
20	550	550	2.44	97.56	85-100
10	20980	21530	95.81	4.19	5-20
4.75	22470	22470	99.92	0.98	0-5

Fig. Results of Individual Gradation of Aggregate Test



Fig. Sieves used for Test

16. Grain Size Analysis Test (IS 2720 - Part 4):

- Objective: To separate the soil into different sizes to see exactly how much gravel, sand, and fine powder (silt/clay) is in it. This helps classify the soil and understand how well it will drain water and carry heavy loads.
- Project Result: A 1000 gm dry soil sample was tested through a series of sieves. The final calculations show the soil is made up of 19% Gravel (retained on the 4.75 mm sieve), 60.9% Sand (between 4.75 mm and 0.075 mm), and 20.1% Silt and Clay (passing the 0.075 mm sieve).
- Interpretation: The results show that almost 80% of the material is coarse (sand and gravel combined). This is an excellent result for road construction. Granular soils like this are very strong, lock together tightly when compacted, and allow water to drain easily, making it a highly stable foundation for the highway.



Fig. Sieves used in Test

		Sole Testing of Special Roadbeds - Shillong/Imphal Road, NEFT Border Greenfield Project (Package 325 Road Km. 53.2-500) (Shillong) to km. 548 and (Imphal) to the state of Manipal/India to be executed on the Road Survey made under the Manipal Performance			
Authority Independent Engineer Consultant/Contractor SPC Contractor		National Highway authority of India Official name: Bastolam JV Highway Roadwork Pvt. Ltd in association with SPC Consultant NHS SH Highway Roadwork Private Limited (CONSOLE) NHS SH India Projects Limited			
GRAIN SIZE ANALYSIS (As per IS 2720 (Part - II))					
Laboratory No.		Date of Sampling			
Type of Material		Date of Testing			
Source		Sampled by			
Location		Tested by			
Prepared Date		Weight of dry sample in grams		1000gms	gms
IS Sieve Size (mm)	Weight of Soil Retained (g)	Cumulative Weight Retained (g)	Cumulative Percentage Retained (%)	Percentage Passing (%)	Remarks
100					Gravel
75					
19	0	0	0	100	
4.75	190	190	19	81	Sand
2.00	41	231	23.1	76.9	
0.425	42.2	652	65.2	34.7	
0.075	146	799	79.9	20.1	Fin & Clay
Pass				20.1	
Remarks:					
Contractor's Representative		Consultant's Representative		Independent Engineer's Representative	

Fig. Results of Grain Size Analysis

17. Specific Gravity and Water Absorption Test (IS 2386 – Part 3):

- Objective: To measure how dense/heavy the stones are (specific gravity) and how much water they soak up (water absorption). Heavier stones are generally stronger, and stones that absorb less water are better because they will not break apart from water damage or soak up too much expensive bitumen during mixing.
- Project Result: Testing the aggregate samples gave the following final results:
20 mm aggregate: Bulk Specific Gravity is 2.943, and Water Absorption is 1.58%.
10 mm aggregate: Bulk Specific Gravity is 2.928, and Water Absorption is 1.85%.
- Interpretation: A specific gravity of 2.943 for 20 mm aggregates and 2.928 for 10 mm aggregates is good (standard road aggregates are usually between 2.6 and 2.9), meaning the material is dense and strong. The standard maximum limit for water absorption in high-quality road surfaces is typically 2%. Since the water absorption for both aggregates is below 2%, it shows the stones are durable and should withstand weather and heavy traffic well.



Fig. Apparatus used for the Test

G R I L NATIONAL HIGHWAYS AUTHORITY OF INDIA			
Project : Se Lining of Surat-Nagahli Ahmednagar-Surat-Mir/NK/ Border Greenfield stretch (Package - XIV from Km 512.000 (Harapur) to km 548.400 (Badadali) in the state of Maharashtra to be executed on Hybrid Annuity Mode under Government Programme.			
Independent Engineer : Global Infra Solutions JV Voyent Solutions Pvt. Ltd in association with M5 Consultant			
Concessionaire : M/s G R Harapur Badadali Highway Private Limited (GRHPL)			
Contractor : G R Infra Projects Limited.			
Determination of Specific Gravity and Water Absorption (IS : 2386 Part-3)			
Description of Materials		Date of Sampling :	
Source Location		Date of Testing :	
Sample at		Sampled By :	
Proposed Use		Tested By :	
		20/07/2019	
		19/07/2019	
Sr. No.	Description	Trial - 1	Trial - 2
1	Weight of Sample in Water W ₁ (gm)	1949	1443.5
2	Weight of SSD Sample W ₂ (gm)	2928.5	2171.5
3	Wt of Oven dry - sample W ₃ (gm)	2885	2132
4	Bulk Specific Gravity, $= W_2 / (W_2 - W_1)$	2.94	2.93
5	Apparent Specific Gravity, $= W_2 / (W_3 - W_1)$	3.03	3.10
6	Water Absorption (%) $= 100 \times (W_2 - W_3) / W_3$	1.58	1.85
7	Average Bulk Specific Gravity		
8	Average Apparent Specific Gravity		
9	Average Water Absorption %		
Remarks :-			
EPC Contractor's Representative			
Concessionaire's Representative			
Independent Engineer's Representative			

Fig. Result from Test

Planning Department

The Planning Department makes sure that all the work on site is running as per the decided schedule in the Concession Agreement. On a large project like this, there are hundreds of activities happening at the same time, so keeping track of everything is very important.

Before the work starts on site, the planning team sits down and breaks the entire project into small individual activities. For road work, they create separate activities like clearing and grubbing, earthwork, granular sub-base laying, prime coat, bituminous layers, and road furniture. For structure work, they break it down activity by activity from foundation level all the way up to the finished structure. Each activity is given a target start date and a target finish date. This complete plan is called the project schedule. I also learned that the more we break down the work into smaller activities, the better we can monitor the project. If an activity has a shorter time span, any delay in it gets noticed very quickly and can be corrected before it becomes a bigger problem. So a detailed breakdown always leads to better control over the project.

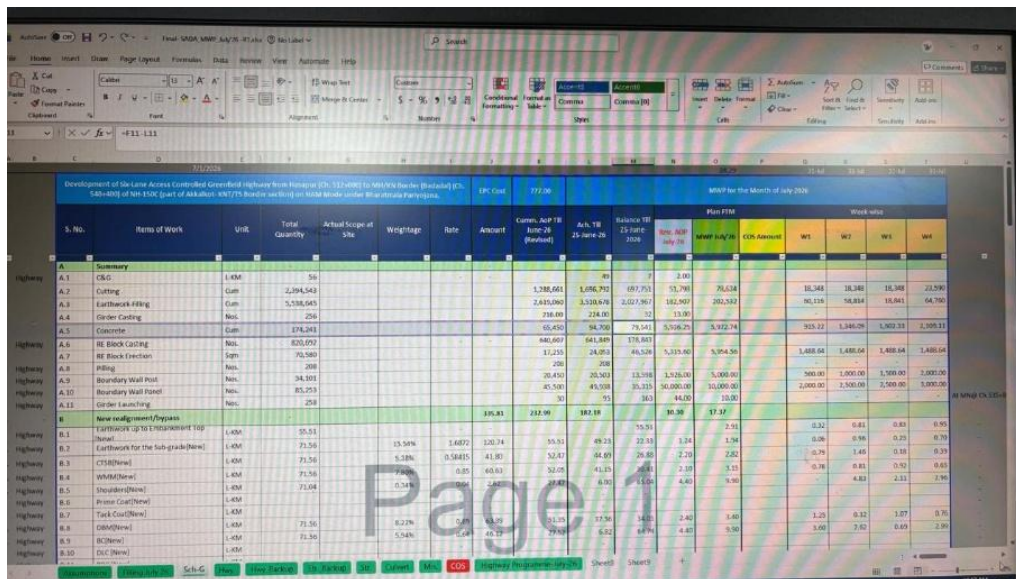
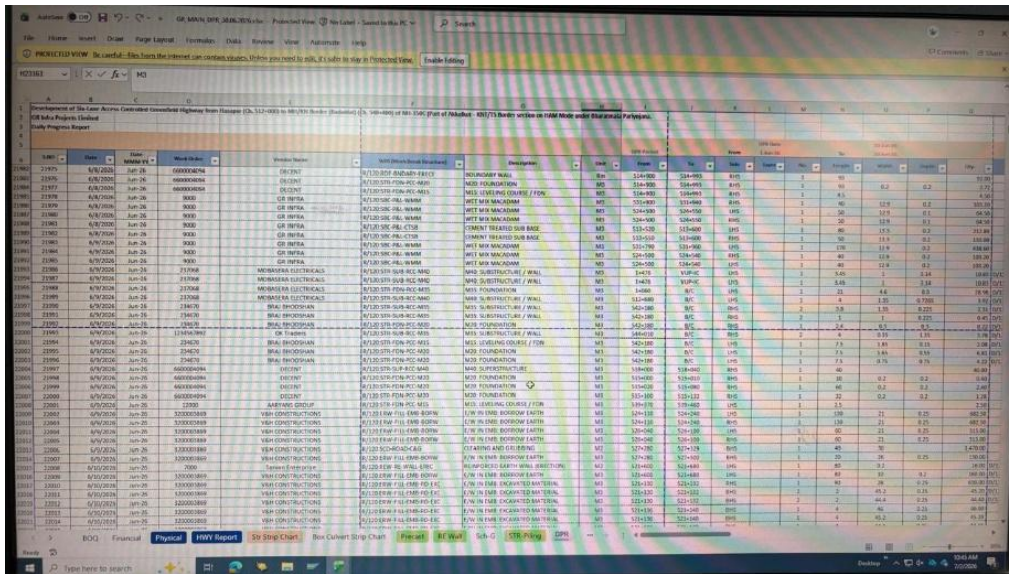
Once work starts on site, the planning team prepares a Daily Progress Report every day, commonly called DPR. For this, they stay in regular contact with site engineers and senior engineers who give them updates on how much work was completed that day. The planning team compiles all this information and compares it with the planned target. If something is running behind schedule, it is flagged immediately so the management can take corrective action.

Another important document I learned about here is the Monthly Progress Report (MPR). This report is prepared every month and shows how much work was completed in that month. It is submitted to NHAI so that they can keep a check on whether the project is on track.

The planning team also prepares a Monthly Work Plan (MWP) at the start of every month. This plan tells the site team which activities will be taken up in the upcoming month, how much material like cement, steel, sand, and bitumen will be required, how many machines will be needed and for how long, and what the target quantities are for each activity. This helps the site team prepare in advance and avoids last-minute shortages of material or machinery.

Another thing I learned was the concept of the Critical Path. In any project, some activities are linked to each other in such a way that if one gets delayed, everything after it also gets delayed. The longest such chain of activities which can not be delayed by single day is called the critical path, and the activities on this chain are called critical activities. These activities have zero float, meaning there is no extra time available even a single day's delay in a critical activity directly pushes the project completion date. Because of this, the planning team keeps a very close watch on critical activities and makes sure they get priority in terms of resources, machinery, and manpower.

All this tracking and scheduling work is done in Microsoft Excel. The team maintains detailed spreadsheets showing activity-wise planned quantities, completed quantities, and balance work. This makes it easy for anyone from the project manager to the client to see the exact status of the project at any point of time.



Billing Department

In the Billing department, I learned how the bills are prepared and how the company gets paid for the work it does. In this project, payment does not happen all at once at the end. Instead, a bill is made after specific percent of work is done. During my internship, there were major fluctuations in the prices of materials, so a monthly billing system was followed for some period of time.

How Bills Are Made

Before any work can be added to a bill, a step called RFI (Request for Inspection) has to be done first. When a piece of work is finished on site like a road layer or a culvert slab the contractor sends an RFI to the Independent Engineer (IE). This RFI is basically a written request saying that a particular work is done and asking the IE to come and check it.

The IE team visits the site, checks the work, and if everything is fine, they approve the RFI. After approval these RFI IDs are then used to prepare the monthly bill. I also got to know that not every RFI is checked by the IE directly. Out of all the RFIs raised, at least 10% are randomly selected and approved by the Independent Engineer after physical site inspection. Another 5% are approved by the Project Director. The remaining RFIs can be approved by the concessionaire itself. This system saves time while still making sure that enough checks are done by the right people.

When Payment is Taken for Different Works

Different types of work have different stages at which payment is taken. The billing team does not bill for every small activity. They bill only when a certain stage is fully completed. Here is how it works:

For Road Work:

For the road, payment is taken after a full pavement layer is completed over a stretch. For example, payment for the sub-base is taken after CTSB or WMM is done. Payment for the surface is taken after DBM or BC (the top layer of the road) is completed.

For Minor Bridges, Major Bridges and Underpasses:

For bigger structures like bridges and underpasses, payment is split into four stages because these structures take a long time to build. The four stages are:

- Stage 1: After the Foundation is done
- Stage 2: After the Substructure (piers and abutments) is done
- Stage 3: After the Superstructure (bridge deck) is done
- Stage 4: After Girder Casting is done

Price Escalation

I also learned about Price Escalation. A highway project runs for 2 or more years. During this time, prices of materials like cement, steel, and fuel keep going up. So the contract has a rule that the payment to the contractor is increased a little based on how much prices have gone up since the day the contract was signed.

In this HAM project, the escalation is calculated using WPI (Wholesale Price Index) with 70% share and CPI (Consumer Price Index) with 30% share. WPI data is taken from the website of the Ministry of Commerce and Industry and CPI data is taken from the website of the Ministry of Labour and Employment. Based on the change in these two indices from the bid date to the invoice date, the extra amount to be paid is calculated and added to the bill.

To find the escalated amount, first a Price Index Multiple is calculated. This is done using the formula:

$$\text{Price Index Multiple} = [(0.7 \times \text{WPI at Invoice Date}) + (0.3 \times \text{CPI at Invoice Date})] \div [(0.7 \times \text{WPI at Bid Date}) + (0.3 \times \text{CPI at Bid Date})]$$

This multiple tells us by how much the prices have gone up from the bid date to the invoice date. The escalated amount is then calculated by multiplying the contract bill amount with this Price Index Multiple. The difference between this escalated amount and the original bill amount is the extra money paid to the contractor to cover the rise in prices.

S. No.	Stage for measurement of Physical Progress	Unit	Qty. (Total)	Weightage Contract Price	Rate	Amount	Upto Date	Upto Previous	This Bill	Upto Date	Upto Previous	This Bill	Remark
1	Road works including culverts, minor bridges, underpasses, approaches to R.O./R.O.D./Major Bridges/ Structures (not including service roads)												
2	New road/repairs	Lm	71.56	15.54%	10,930,729	3,355,338,740	45,872	41,942	1,930	868,760,120	832,308,258	36,551,862	
3	Earthwork up to top of the sub-grade	Lm	71.56	5.38%	6,556,979	468,344,730	41,731	39,604	2,127	273,629,589	259,682,594	13,946,994	
4	Transfer work (sub-base, base, shoulder)	Lm	71.56	7.80%	9,510,291	680,596,075	41,323	39,454	2,907	392,974,719	365,338,304	27,646,415	
5	WMM	Lm	71.04	0.34%	413,693	29,390,493							
6	Shoulders	Lm	71.56	8.22%	10,020,763	717,127,642	37,504	30,991	6,513	375,818,714	310,553,481	65,265,233	
7	Bituminous work	Lm	71.56	5.94%	7,233,795	517,680,522	6,300	6,300		45,572,906	45,572,906		
8	Rigid Pavement	Lm		0.00%									
9	Concrete work	Lm											
10	Culverts	No.	212.00	7.73%	3,102,664	674,512,808	151,000	149,000	2,000	480,431,292	474,067,964	6,363,328	
11	Minor bridges	No.	38.00	2.29%	5,799,518	19,608,242	14,000	14,000		185,378,393	185,378,393		
12	Foundation	No.	60.00	2.01%	2,925,000	17,550,000	47,000	47,000		142,844,790	137,014,391	5,830,400	
13	Sub-structure	No.	28.00	0.00%									
14	Super structure (including crash barriers etc. complete)	No.	64.00	0.57%	780,857	49,914,048	60,000	60,000		39,042,855	39,042,855		
15	(On Order Casting 100%)	No.	28.00	0.86%	2,677,224	74,962,281	10,000	9,000	1,000	28,772,243	24,095,019	2,677,224	
16	On Completion (50%)	No.											
17	Cattle/human/underpasses	No.											
18	Foundation	No.											
19	Sub-structure	No.											
20	Super structure (including crash barriers etc. complete)	No.											
21	Pedestrian overpasses	No.											
22	Foundation	No.											
23	Sub-structure	No.											
24	Super structure (including crash barriers etc. complete)	No.											

Fig. Billing Sheet as per Schedule-G of the Concession Agreement

Escalation											
Name of Project: Development of Six Lane Access Controlled Greenfield Highway from Hasapur (Ch. 512+000) to MIDPN Border (Badaldal) (Ch. 548+000) of NH 150C (Part of Akkalkot - KNTI) S Border section on HAM Mode under Bharatmala Pariyojana.											
Bid Date		3/13/2023		Date of Invoice:				12/25/2025			
Bid Project Cost (A)		8,721,700,000		Date Price Index Preceding the Bid date							
Appointed date		10/3/2024									
				WPI		CPI (IW Solapur)		Price Index Preceding the Bid Date			
Ref. Date		Index		Ref. Date		Index					
Feb/23		150.90		Dec/22		137.90		147.06			
Adjustment of Bid Project Cost with Price Index Multiple (Cl. 23.2 of CA)											
S.No.	Payment Milestone Ref.	Invoice Date	WPI (All Commodities)		CPI (IW Solapur)		Price Index on date of invoice as per model circular dated 02.12.2020	Price Index Multiple (PIM)	Adjusted Bid Project Cost	Payable Amount 4% of the Adjusted Project Cost	Remarks
			Ref. Date	Index	Ref. Date	Index					
a	b	c	d	e	f	g	h=ex70%+gx30%	i	k	l	i
1	Payment Milestone - 01	22/Apr/25	Mar/25	154.89	Mar/25	152.30	154.0500	1.050	9,157,785,000	366,311,400	PMS-I
2	Payment Milestone - 02	09/Jun/25	May/25	153.78	Mar/25	152.30	153.2000	1.040	9,070,568,000	362,822,720	PMS-II
3	Payment Milestone - 03	12/Sep/25	Aug/25	155.28	Jun/25	152.30	154.3000	1.050	9,157,785,000	366,311,400	PMS-III
4	Payment Milestone - 04	06/Jan/26	Dec/25	157.20	Dec/25	153.90	156.2100	1.060	9,245,002,000	369,800,080	PMS-IV
5	Payment Milestone - 05	14/Mar/26	Feb/26	158.49	Dec/25	153.90	157.0500	1.070	9,332,219,000	373,288,760	PMS-V
6	Payment Milestone - 06	31/May/26	Apr/26	167.00	Mar/26	155.90	163.6700	1.110	9,681,087,000	387,243,480	PMS-VI
ABS_Top Escalation Cost Escalation-Compensation Material Consumption Schedule G -SADA Subgrade CTSBGSB WMM DBM Road side Drain Box Culvert Underpass Foundation											

Fig. Escalation Calculation Sheet

Safety Department

In the Safety department, I did not go to the construction site for an inspection. Instead, the safety officers sat with us and explained all the important safety rules, equipment, and procedures that are followed on a project like this. Even though it was a classroom-style session, I learned a lot of practical things that are directly used on site every day. Also learned about the sign boards that are used at site during work around the traffic.

PPE (Personal Protective Equipment)

The first thing the safety team explained was PPE, which stands for Personal Protective Equipment. On a construction site, every single person whether they are a worker, an engineer, or a visitor must wear proper PPE before entering the work area. The PPE kit includes:

- Safety helmet: to protect the head from falling objects or accidental hits
- Safety shoes: to protect feet from heavy materials or sharp objects on the ground
- Safety jacket or high-visibility vest: so that workers can be easily seen by machine operators and other workers on a busy site
- Safety gloves: to protect the hands while handling steel bars, concrete, or chemicals
- Safety goggles: to protect the eyes from dust, sparks, or flying debris
- Dust mask: to protect the lungs from breathing in harmful dust
- Ear plugs: to protect the ears from damage due to loud noise from heavy machines

Work Permits for Risky Activities

Another very important thing I learned in this session was about Work Permits. On a construction site, not every job can be started without permission. For activities that are considered risky or dangerous, the worker or supervisor must first get a written permit from the safety department. This permit is called a PTW (Permit to Work). The safety team explained that the following types of permits are issued on this site:

- Electrical Permit: Required before doing any work involving electricity or electrical equipment
- Height Permit: Required before working at heights, for example when workers go up on scaffolding, formwork, or a bridge structure
- Excavation Permit: Required before digging or excavating the ground, especially for excavations that are 1.2 metres or deeper, because deep pits can collapse and be very dangerous
- General Permit: Required for general risky activities that do not fall under the above specific categories
- Critical Lifting Permit: Required for heavy lifting activities like girder launching, where large cranes are used to lift massive precast girders and place them on bridge piers. This

is one of the most dangerous activities on the site and needs extra safety checks before it can begin

LOTO (Lockout Tag Out)

The safety team also explained a procedure called LOTO, which stands for Lockout Tag Out. This procedure is followed whenever maintenance or repair work is being done on a machine or plant like the RMC (Ready Mix Concrete) plant to make sure that the machine cannot be accidentally switched on while someone is working on it. A physical lock is put on the machine's power switch and a tag is attached to it saying that it is under maintenance. Only the person who put the lock is allowed to remove it. This is done to prevent serious accidents.

Safety Induction

The safety team also told us about a Safety Induction that every new worker or visitor must attend before stepping on the site. This induction covers the basic rules of the project site, the location of first aid boxes, emergency contacts, and how to behave safely around machines and work zones.



Fig. PPE Kit



Fig. Safety Signs

Plants

On this project, there are several plants set up near the site. I visited them and got to see how each one works. Below is a description of each plant along with photos I took during my visit:

1. Hot Mix Plant

The Hot Mix Plant is where bitumen and aggregates (stones and sand) are heated and mixed together to produce the asphalt mix that is used to lay the road surface. The process starts with raw aggregates being fed into a rotating drum dryer which removes all moisture from them. The hot dry aggregates are then mixed with hot liquid bitumen in the correct proportions as per the mix design. The final hot mix is loaded into trucks and immediately taken to the paving site, where it is spread by a paver machine and compacted by rollers. The temperature of the mix must be maintained throughout if it cools down too much before laying, it cannot be compacted properly.



Fig. Hot Mix Plant

2. RMC Plant (Ready Mix Concrete Plant)

The RMC Plant is used to produce concrete that is used in all the structural work on the project like bridges, culverts, piers, pier caps, box culverts, and buildings. At the RMC plant, cement and fly ash (stored in large tall silos), sand, aggregates, water, and admixtures are all measured precisely and mixed together in the right proportions as per the concrete mix design. The fresh concrete is then loaded into transit mixer trucks (the drums that rotate continuously while moving) and delivered to the structure site.



Fig. RMC Plant

3. Aggregate Crushing and Screening Plant

This plant is where large stones (boulders) are crushed down into smaller aggregate sizes that are required for concrete and road construction. The crushed material then passes through a series of vibrating screens which separate the aggregates into different sizes for example 10mm, 20mm, and 40mm. Each size goes into a separate stockpile and is used for different purposes in the project. This plant makes sure that a continuous supply of properly sized aggregates is available on site without depending entirely on outside suppliers.



Fig. Aggregate Crushing and Screening Plant

4. Sand Washing Plant

The sand washing plant is used to clean and process sand before it is used in concrete or road construction. The raw sand dug from the ground often contains silt, clay, dust, and other fine particles that reduce the quality of concrete. This plant washes the sand with water and removes all the unwanted fine material, giving a clean graded sand that meets the required quality standards. Good quality washed sand is very important because it directly affects the strength and durability of the concrete used in the structures.



Fig. Sand Washing Plant

5. RE Block Plant

RE blocks are small concrete blocks used to build retaining walls on the sides of the highway. These walls stop the soil from sliding down where the road is at a higher level than the ground around it.

At our project site, these blocks were made at the casting yard itself. The plant has a cement silo, aggregate bins, a conveyor belt, and a pressing machine. All the materials are put into the machine, which presses them into a block shape. After that, the blocks are kept for curing for a few days. Before sending them to site, quality checks are done to make sure the blocks are strong enough.



Fig. RE Block Plant

Structure Department

The Structure department was one of the most interesting parts of my internship. This department takes care of all the bridges, culverts, underpasses, interchanges, and buildings that are built along the highway. A highway is not just one long road it has many different structures built at various points to handle traffic crossings, drainage, and railway lines. During my time in this department, I went on several site visits and saw almost all types of structures present in this project, and I also looked at their construction drawings to understand them properly.

Types of Structures I Saw on Site

- Girder:- A girder is a large, long concrete or steel beam that is placed horizontally between two piers (pillars) on bearing pedestals to support the bridge deck above it. In this project, precast post tension girders are cast separately and then lifted and placed on top of bearing pedestals using cranes. This activity is called girder launching, and it is one of the most critical and risky jobs on a bridge site because the girders are extremely heavy.



Fig. Post Tensioning of Girder

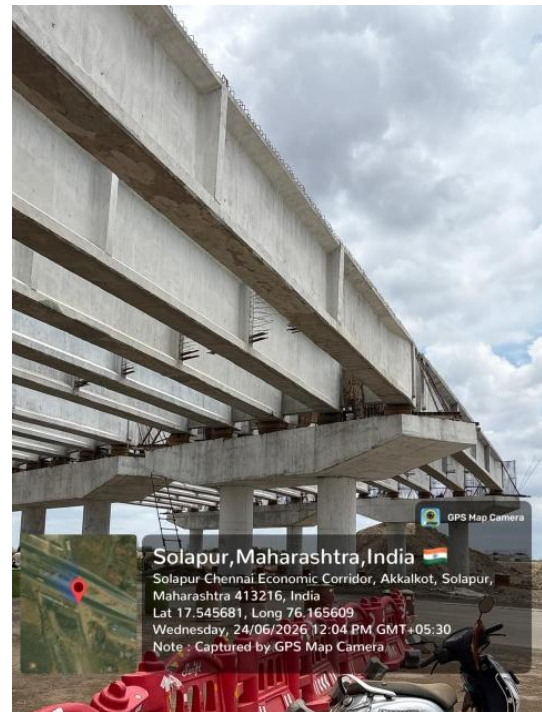


Fig. Girder Placed on the Bearing Pedestal



Fig. Precast Post tensioned girder kept in casting Yard

- VUP (Vehicular Underpass):- A VUP is a small tunnel-like structure built below the highway so that local traffic, especially smaller vehicles, two-wheelers, and farm vehicles, can cross from one side of the highway to the other without climbing up to the highway level. This project has 2 VUPs and 9 LVUPs (Light Vehicular Underpasses).



Fig. VUP at Interchange

- MNB (Minor Bridge):- A minor bridge is a smaller bridge (generally of length 6 m to 60 m) built over small rivers, nallahs (drains), or low-lying areas where water collects during the monsoon. This project has 13 minor bridges along its 36.4 km length.



Fig. Minor Bridge Construction is in progress

- **MJB (Major Bridge):-** A major bridge is a bigger structure (generally of length greater than 60 m), usually built over a wider river or a larger water body. It needs deeper foundations and stronger piers compared to a minor bridge. This project has 1 major bridge.



Fig. Major Bridge Under Construction

- **Box Culvert:-** A box culvert is a box-shaped structure (generally of length less than 6 m), usually made of reinforced concrete, that is built under the road to allow rainwater or

small streams to pass from one side to the other. This project has 96 box culverts and 10 pipe culverts, which shows how important drainage is for a highway of this size.



Fig. Box Culvert

- RE Wall: An RE wall is a vertical retaining wall built at the approaches of bridges and underpasses to securely hold the high soil embankments in place. In this project, it is constructed using precast RE blocks on the outside face. Right behind these blocks, a layer of filter media (like crushed stone or gravel) is placed to allow proper water drainage, and after that, the main soil is filled and compacted. Geogrids are tied to the blocks and buried tightly within this soil to give the wall its strength. This vertical design saves significant land space by avoiding the need for wide, sloping soil banks.

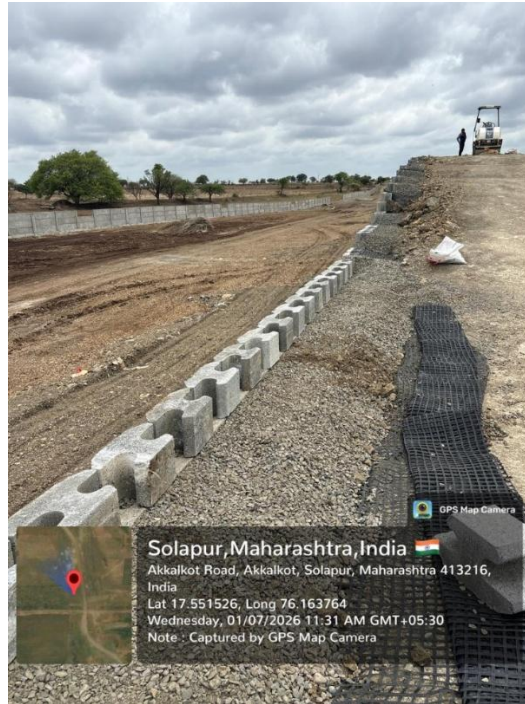


Fig. RE Wall under Construction

- **Building:-** Apart from bridges, culverts, and underpasses, the project also includes the construction of several buildings such as the Admin Building, Medical Block, Toilet Block, and Highway Nest. These buildings are smaller in size compared to bridges, but they are still designed using proper structural drawings, with their own foundation, column, and roof slab details, just like any other RCC (Reinforced Cement Concrete) structure.



Fig. Medical Block work in Progress

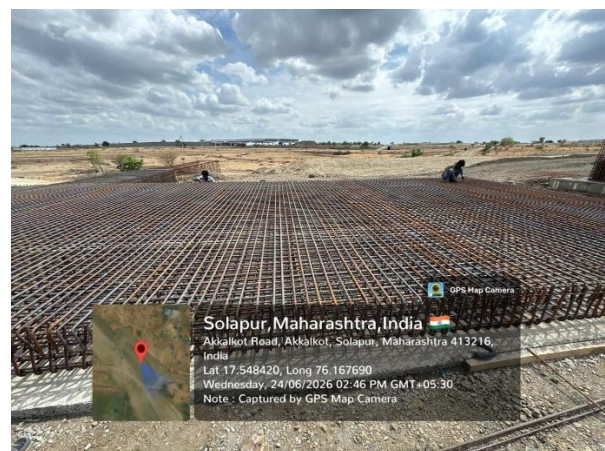


Fig. Admin Block Bars for Raft being placed

Understanding Structural Drawings

For each of the structures mentioned above, I was shown the construction drawings on site and was able to compare them with the actual structure being built. A structural drawing shows the exact shape, size, dimensions, and reinforcement details of a structure. Reading these drawings is an important skill because it lets an engineer check whether the work happening on site is exactly matching what was designed.

I looked at drawings for all the structure types like girders, VUPs, minor bridges, major bridge, box culverts, interchange, and the admin building and understood how each one is detailed differently depending on the load it has to carry and the span it has to cover.

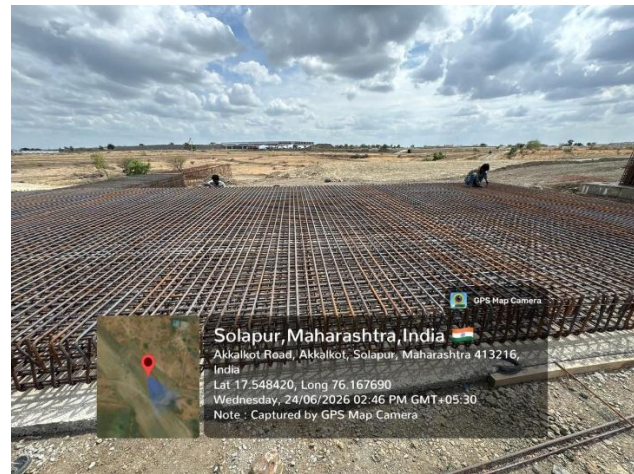
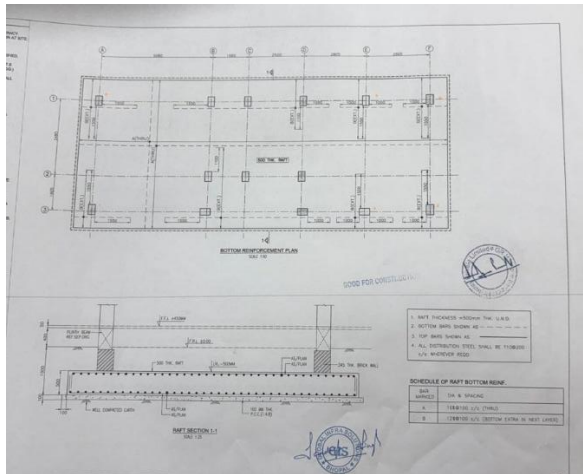


Fig. Site execution as per drawing for Admin Building

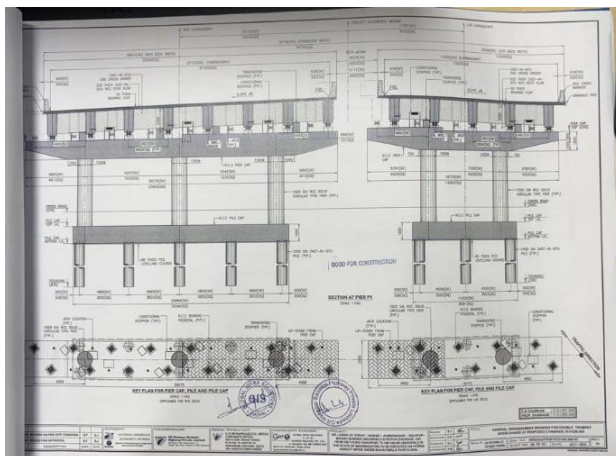


Fig. Site execution as per drawing for Double Trumpet Interchange

One of the things I observed closely in this department was the Bar Bending Schedule, commonly called BBS. A BBS is basically a detailed list that is prepared from the structural drawing. It tells the exact diameter of each steel bar (like 12mm, 16mm, 20mm, 25mm), the length each bar needs to be cut to, the shape it needs to be bent into (straight, L-shaped, U-shaped, etc.), and the total number of such bars required for that particular structural member. I prepared the BBS myself for some of the building parts like beams, columns, etc. For other structures I studied the already made BBS and I observed how the site engineers use the structural drawing to work out the BBS for different members like pier caps, girders, slabs, and box culverts. This was very useful because it showed me how a 2D drawing on paper is converted into a practical.

I also understood why BBS is so important, without it there would be a lot of wastage of steel, the bars might be cut to the wrong length, or the wrong shape might be used in the wrong place. A proper BBS makes sure that steel ordering, cutting, and bending all happen accurately, which saves both material and money on a big project like this.

Station	No. of Bars	Bar Number Description	Slope	Diameter of Bar (mm)	Length (m)	Type	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	aa	ab	ac	ad	ae	af	ag	ah	ai	aj	ak	al	am	an	ao	ap	aq	ar	as	at	au	av	aw	ax	ay	az	ba	bb	bc	bd	be	bf	bg	bh	bi	bj	bk	bl	bm	bn	bo	bp	bq	br	bs	bt	bu	bv	bw	bx	by	bz	ca	cb	cc	cd	ce	cf	cg	ch	ci	cj	ck	cl	cm	cn	co	cp	cq	cr	cs	ct	cu	cv	cw	cx	cy	cz	da	db	dc	dd	de	df	dg	dh	di	dj	dk	dl	dm	dn	do	dp	dq	dr	ds	dt	du	dv	dw	dx	dy	dz	ea	eb	ec	ed	ee	ef	eg	eh	ei	ej	ek	el	em	en	eo	ep	eq	er	es	et	eu	ev	ew	ex	ey	ez	fa	fb	fc	fd	fe	ff	fg	fh	fi	fj	fk	fl	fm	fn	fo	fp	fq	fr	fs	ft	fu	fv	fw	fx	fy	fz	ga	gb	gc	gd	ge	gf	gg	gh	gi	gj	gk	gl	gm	gn	go	gp	gq	gr	gs	gt	gu	gv	gw	gx	gy	gz	ha	hb	hc	hd	he	hf	hg	hh	hi	hj	hk	hl	hm	hn	ho	hp	hq	hr	hs	ht	hu	hv	hw	hx	hy	hz	ia	ib	ic	id	ie	if	ig	ih	ii	ij	ik	il	im	in	io	ip	iq	ir	is	it	iu	iv	iw	ix	iy	iz	ja	jb	jc	jd	je	jf	jg	jh	ji	jj	jk	jl	jm	jn	jo	jp	jq	jr	js	jt	ju	jv	jw	jx	jy	jz	ka	kb	kc	kd	ke	kf	kg	kh	ki	kj	kk	kl	km	kn	ko	kp	kq	kr	ks	kt	ku	kv	kw	kx	ky	kz	la	lb	lc	ld	le	lf	lg	lh	li	lj	lk	ll	lm	ln	lo	lp	lq	lr	ls	lt	lu	lv	lw	lx	ly	lz	ma	mb	mc	md	me	mf	mg	mh	mi	mj	mk	ml	mm	mn	mo	mp	mq	mr	ms	mt	mu	mv	mw	mx	my	mz	na	nb	nc	nd	ne	nf	ng	nh	ni	nj	nk	nl	nm	nn	no	np	nq	nr	ns	nt	nu	nv	nw	nx	ny	nz	oa	ob	oc	od	oe	of	og	oh	oi	oj	ok	ol	om	on	oo	op	oq	or	os	ot	ou	ov	ow	ox	oy	oz	pa	pb	pc	pd	pe	pf	pg	ph	pi	pj	pk	pl	pm	pn	po	pp	pq	pr	ps	pt	pu	pv	pw	px	py	pz	qa	qb	qc	qd	qe	qf	qg	qh	qi	qj	qk	ql	qm	qn	qo	qp	qq	qr	qs	qt	qu	qv	qw	qx	qy	qz	ra	rb	rc	rd	re	rf	rg	rh	ri	rj	rk	rl	rm	rn	ro	rp	rq	rr	rs	rt	ru	rv	rw	rx	ry	rz	sa	sb	sc	sd	se	sf	sg	sh	si	sj	sk	sl	sm	sn	so	sp	sq	sr	ss	st	su	sv	sw	sx	sy	sz	ta	tb	tc	td	te	tf	tg	th	ti	tj	tk	tl	tm	tn	to	tp	tq	tr	ts	tt	tu	tv	tw	tx	ty	tz	ua	ub	uc	ud	ue	uf	ug	uh	ui	uj	uk	ul	um	un	uo	up	uq	ur	us	ut	uu	uv	uw	ux	uy	uz	va	vb	vc	vd	ve	vf	vg	vh	vi	vj	vk	vl	vm	vn	vo	vp	vq	vr	vs	vt	vu	vv	vw	vx	vy	vz	wa	wb	wc	wd	we	wf	wg	wh	wi	wj	wk	wl	wm	wn	wo	wp	wq	wr	ws	wt	wu	wv	ww	wx	wy	wz	xa	xb	xc	xd	xe	xf	xg	xh	xi	xj	xk	xl	xm	xn	xo	xp	xq	xr	xs	xt	xu	xv	xw	xx	xy	xz	ya	yb	yc	yd	ye	yf	yg	yh	yi	yj	yk	yl	ym	yn	yo	yp	yq	yr	ys	yt	yu	yv	yw	yx	yy	yz	za	zb	zc	zd	ze	zf	zg	zh	zi	zj	zk	zl	zm	zn	zo	zp	zq	zr	zs	zt	zu	zv	zw	zx	zy	zz
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Fig. BBS made as per Drawing for Plinth Beam of Admin Building

Survey Department

Survey is one of the most important departments on a highway project. Without accurate survey work, nothing on site can be built at the correct location, level, or alignment.

I learned that before the actual construction starts, NHAI gives the concessionaire some fixed reference points. These points are placed at the median and at the end of the Right of Way at every 5 km interval. Each of these points has a known coordinate and altitude. The survey team then uses these reference points as a base and works outward from them. Using a Total Station, they find the coordinates and altitude of several other points along the entire stretch at different distances depending on the curves and slopes of the ground. All these collected points are then plotted in AutoCAD to make a drawing that shows the exact ground conditions of the site.

Once construction starts, this pre-collected data becomes very useful. Whenever any structure or road layer is being built, the survey team uses the Total Station again to check whether it is being constructed at the correct location, alignment, and level as per the drawing. If anything is even slightly off, it is corrected immediately before the work goes further.

For level-related work like filling of material in embankment or checking the thickness of road layers, the survey team uses an Auto Level. It is simpler and faster to use than a Total Station and is very effective where only height or level needs to be checked.



Fig. Total Station Mounted on the Stand and Prism placed on Rod

Highway Department

The Highway Department handles all the road construction work from the preparation of the ground at the bottom all the way up to the final road surface on top. The road constructed on this project is a flexible pavement. In a flexible pavement, the load of the vehicles is transferred layer by layer from the top surface down to the ground below. Each layer has a different material and purpose, and together they distribute the load safely to the subgrade.

The first thing I learned here was about the Typical Cross Sections (TCS) of the road. A cross section is a drawing that shows how the road looks if you cut it from one side to the other.

Different locations on the project have different cross sections depending on the ground condition and road requirement. For example, on our project I studied cross sections like a 6-lane divided highway with flush median, a 6-lane highway in rock cutting, a section near the VOP approach with slip roads, a 4-lane section at the interchange loop, and a 6-lane section with RE wall on the highway shoulder. Each of these has different widths and arrangements but the pavement layers and slopes remain mostly the same across all sections the carriageway has a cross slope of 2.5% and the shoulders have a slope of 3.0%. The pavement layers from bottom to top are Subgrade, CTSB, Geogrid, WMM, DBM, and BC.

I also visited the site and observed the actual construction work. I saw how the Subgrade is prepared by shaping and compacting the natural soil to the required level and slope. After that, each layer of material is laid one by one and compacted properly before the next layer is placed on top.

I also got to see DBM (Dense Bituminous Macadam) laying on site. DBM is a bituminous layer that is laid using a paver machine. After laying it is compacted with help of Rollers and then, a PTR (Pneumatic Tyre Roller) is used to press the material and fill any small gaps, giving a smooth and dense surface. Compaction is very important at every stage because a poorly compacted layer can lead to cracks and damage in the road later.

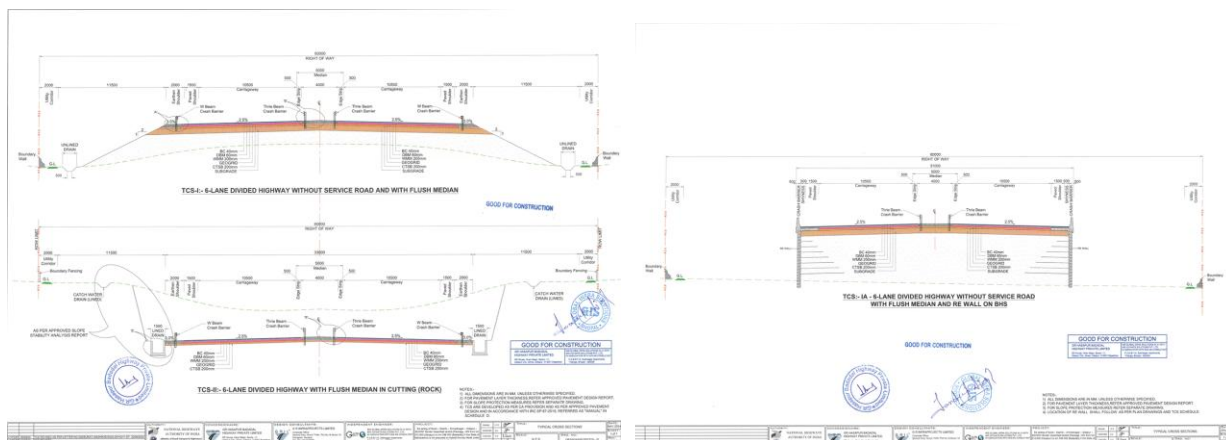


Fig. Typical Cross Sections



Fig. DBM Laying



Fig. Compaction of DBM after Laying



Fig. PTR using on DBM Layer

Store and Purchase Department

The Store and Purchase Department is responsible for making sure that everything needed on site and at the camp is available at the right time. This department does not just handle construction materials it takes care of everything from cement and aggregate to vehicle spare parts, PPE kits, coolers in rooms, and any other item that is needed on site or at the camp.

I learned that there is a proper process for ordering anything on site. Whenever someone needs something, they cannot just go and take it directly. First, they have to raise a Purchase Requisition (PR). This PR then goes to the manager of that department for approval. After the department manager approves it, it goes to the Project Manager. If the Project Manager also approves it, it is then forwarded to the higher authority. The higher authority reviews it and decides whether the item actually needs to be ordered or not. Only after all these approvals does the Store and Purchase Department place the order.

Once the material arrives on site, the store team does not hand it over immediately. They first verify the received material against the purchase invoice to make sure that what was ordered is exactly what has been delivered correct quantity, correct specification. After verification, they inform the person who raised the PR that the material is ready to be collected.

However, this full approval process is not needed for everything. Items that are regularly used on site like PPE, couplers, bolts, and similar small things are already kept in stock by the store team. For urgent situations, the concerned managers directly contact the higher authority and the order is placed immediately without waiting for the full process.

For construction materials like cement, aggregate, steel, and bitumen, the orders are placed based on the Monthly Work Plan (MWP) prepared by the Planning Department. This way, the materials reach the site just when they are needed, and there is no unnecessary stockpiling or shortage.



Fig. Things Put in Store